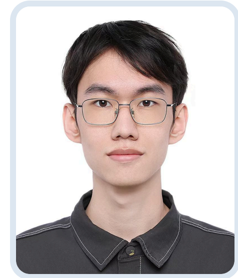
Xinyu Chen

Beijing, China chenxinyu20040529@gmail.com +86 18911827529



Education

Peking University, Tong Class, Yuanpei College Sept 2022 - Jul 2026

- **Major:** Artificial Intelligence
- **GPA:** 3.565/4.0 | 85/100 (first three years: 3.667/4.0 | 86.7/100)
- **Core Courses:** Machine Learning (91), Mathematical Foundation for AI (90), Multi-agent System (94), Introduction to Computer Vision (93), AI Systems Practice I (96), AI Systems Practice II (97), Cognition and Reasoning (90.5), Deep Learning and Natural Language Processing (93)
- Member of PAIR-Lab and intern at BAAI

Peking University, School of Computer Science, Ph.D. Student Sept 2026 - Present

- **Advisor:** Tengjiao Wang

English

CET-4: 605 **CET-6:** 664 **TOEFL:** 107

Research Experience

AI for Law, Peking University / BIGAI Mar 2024 - Jun 2024

- Constructed a dataset for structured extraction of key legal case elements.
- Demonstrated that fine-tuned large language models can extract critical features of legal cases to support model-based case judgments.

Safety Evaluation of Large Language Models, PAIR Lab Sept 2024 - Nov 2024

- Explored the DNA paradigm for category-agnostic and scalable all-round safety evaluation of large language models.

LLM & VLM Alignment, PAIR Lab Jun 2024 - May 2025

- Contributed to the *Align Anything* framework, including training code, interactive terminals, and datasets.
- Developed the Beaver-V preference dataset to improve the safety and helpfulness of vision-language models.
- Designed the Safe RLHF-V algorithm to balance VLM safety and helpfulness.

LLM-based Social Relation Understanding, Peking University / BIGAI Dec 2024 - May 2025

- Collaborated with Fanqi Kong to explore adversarial model generation methods for improving bot detection in large language models.

LLM Math Agent Sept 2025 - Present

- Working on automated mathematical proof verifiers based on step decomposition and fine-grained review.

Multi Agent LLM Sept 2025 - Present

- Working on multi-agent reinforcement learning methods to improve long-context capabilities of language models.

LLM LoRA Memory May 2026 - Present

- Developing new methods to substantially improve model capabilities during test-time training.

AI for Perovskite Sept 2025 - Present

- Using fine-tuned pretrained molecular models to predict delta PCE for dopant molecules in perovskite solar cells.

AI for Cryo-ET Reconstruction Sept 2025 - Present

- Building a variational-inference-based Cryo-ET reconstruction paradigm and demonstrating its usefulness and inspiration for the community from both theoretical and experimental perspectives.
- Developed a series of works including CryoGEN, CryoGEN-II, and CryoWGEN; completed the Cryo-ET blog and the knowledge base Open Cryo-ET.

Research Outputs

Safe RLHF-V: Safe Reinforcement Learning from Multi-modal Human Feedback NeurIPS 2025

CryoGEN-II: Cryogenic Electron Tomography Reconstruction via Generative Network CVPR 2026 Workshop

EVIA: Entropic Variational Inference Auto-encoding UAI 2026

Enhancing LLM-Based Social Bot via an Adversarial Learning Framework EMNLP 2025 Oral

SIV-Bench: A Video Benchmark for Social Interaction Understanding and Reasoning ACL 2026

Reinforced Self-play for Long-Context Understanding In preparation

Veri Graph: A Graph-Based Agentic Workflow for Mathematical Proof Verification In preparation

Transfer learning discovery of modulating molecules for perovskite solar cells In preparation

Contest

Innovation Category at the 9th National Youth AI Innovation and Entrepreneurship Conference Dec 2024

Team Leader of Zhichi, First Prize

- Proposed an ensemble-learning model to predict the spatial relationship between the third molar and mandibular canal, surpassing professional dental experts.
- Designed the system to assist clinical dentists in preoperative diagnosis for wisdom tooth surgery.

Projects

Align-Anything repo

- Implemented multiple LLM alignment algorithms, including ORPO, SimPO, KTO, and PEFT algorithms.
- Led development of the interactive terminal.
- Contributed to the align-anything dataset.

Beaver-V repo

- Implemented code for Safe-RLHF-V.
- Constructed the multimodal preference dataset Beavertails-V.

AgentMind Agent platform

Full-process agent platform for data collection, knowledge-base RAG graph construction, and deep research.

- Supports multi-source data collection and research orchestration, integrating web search, Baidu Hot Search, arXiv, webpage crawling, planning, analysis, and report generation.
- Supports knowledge-base RAG and entity-enhanced graph construction, including document chunking, vector retrieval, LinearRAG entity extraction, entity statistics, and entity-aware recall.
- Supports deep research agent workflows across planning, collection, analysis, verification, and report generation, producing structured reports with citations, charts, and conclusions.
- Collaborated with Costar Group for deployment and migration, serving its optical large-model dialogue platform.

Implementation of LLM-based Multimodal Semantic Compression Undergraduate Thesis

- Completed and defended the thesis, receiving an **Excellent** evaluation.

Talks

- CryoGEN-II: Cryogenic Electron Tomography Reconstruction via Generative Network

Selected Awards

- **2024:** Xingzhengde Scholarship (¥5000 RMB)
- **2023:** Peking University Social Work Award
- **2022:** Linhu Scholarship (¥50000 RMB)

Languages

- **Programming Languages:** C++, C, CUDA, Python
- **Natural Languages:** Chinese (Native), English (Proficient)

Blogs

- Cryo-ET blog